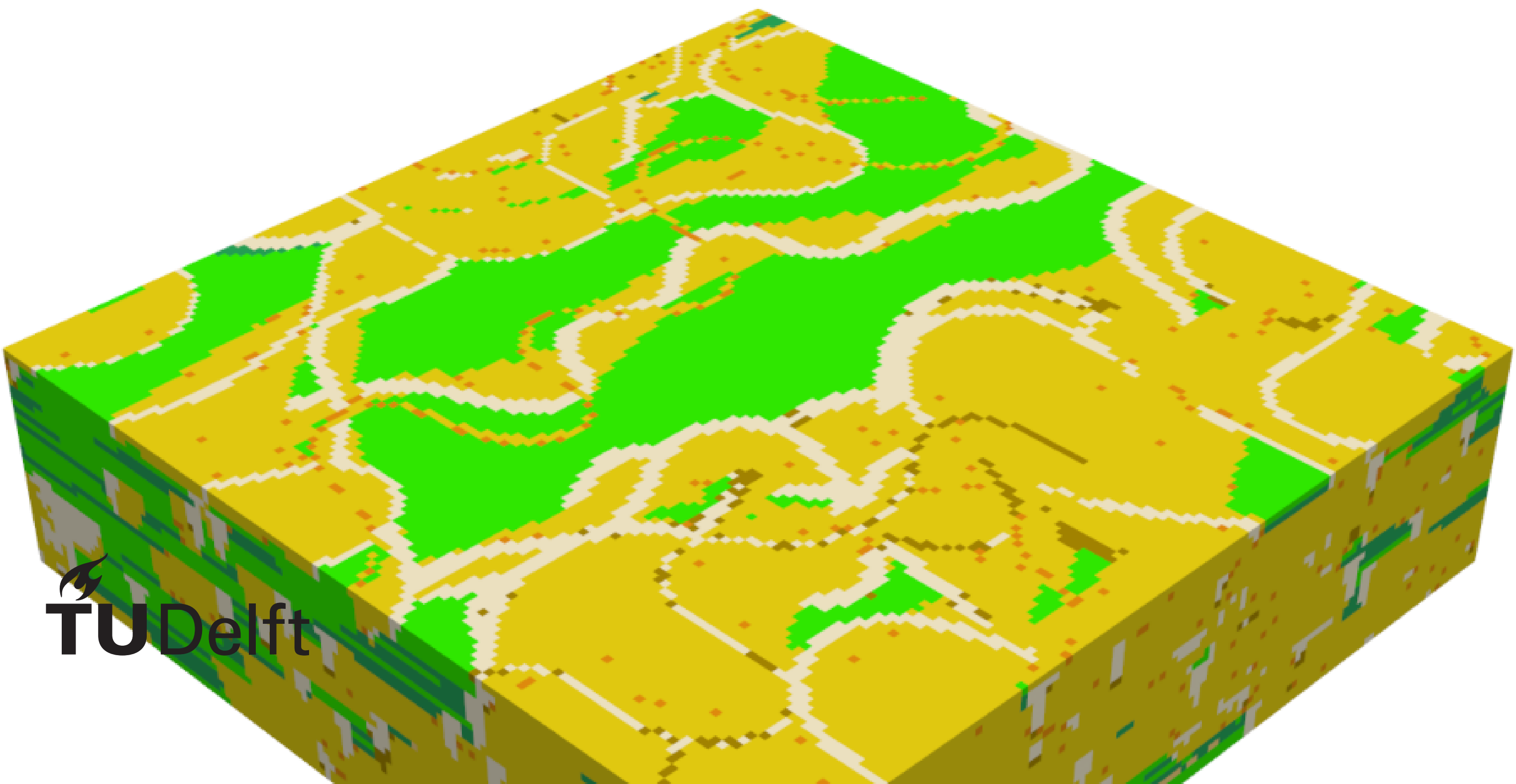
Evaluating the feasibility of Generative Adversarial networks to generate geological facies models

A Case study for the Delft Area

M.D. Torres Ruhe



TU Delft

Evaluating the feasibility of Generative Adversarial networks to generate geological facies models

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by

M.D. Torres Ruhe

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Abstract

Here is the text of my abstract...

Acknowledgements

I would like to thank...

List of Abreviations

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cGAN Conditional Generative Adversarial Network. 11

DSST Delft Sandstone. 3, 4

FluvGan Fluvial Generative Adversarial Network. 11

GAN Generative Adversarial Network. 11

TUD Technische Universiteit Delft. 3

TvD True Vertical Depth. 3

WNB West Netherlands Basin. 3

Contents

1 Introduction

Sedimentary deposits represent a vital commodity in modern society due to the various resources that can be extracted from them. These resources include minerals, groundwater, geothermal heat, sands and hydrocarbons (e.g., Brekke et al., 2001; Donselaar et al., 2017; Donselaar et al., 2015; Morris & Ramanaidou, 2007). Moreover, recent studies have investigated storage possibilities for carbon dioxide and hydrogen in this subsurface environment (Heinemann et al., 2021; Zhou et al., 2020). Beyond the increase in subsurface applications, the world also deals with an increasing population and energy demand. Consequently, the demand for resources hosted within fluvial sedimentary deposits is expected to increase while the distribution of the commodities themselves becomes sparser and of lower quality (e.g., Calvo et al., 2016; Spittler et al., 2020). This highligts the need for a deeper understanding of subsurface environments to enable more precise resource management which can be achieved through various methods such as: data gathering surveys, geological evolution and constructing high-quality 3D geomodels.

3D geomodels are traditionally constructed by engineers and geoscientists to predict reservoir properties and resource distribution (Royer et al., 2015; SONG et al., 2022). These predictions are fundamentally based on a foundational geological conceptual model. The conceptual model acts as a hypothesis regarding the most likely rock mass configuration, built entirely from sparse field data (Koltermann & Gorelick, 1996). This data sparsity is an unavoidable reality of subsurface exploration, resulting from the exceptionally high costs and extended runtimes required for physical data acquisition. Consequently, engineers must develop a working understanding of the subsurface based on highly limited information. This reliance on limited information drives the continuous need to improve geomodelling techniques. Such improvements aim to computationally bridge the gap between isolated data points and realistic, continuous geological representations.

1.1. Problem definition

Despite ongoing advancements in geomodelling techniques, the subsurface environment remains inherently characterized by a lack of data. This data scarcity dictates that functional 3D geomodels must rely heavily on subjective interpretations of the underlying geological conceptual model. Because transitioning from these interpretive hypotheses into a digital framework requires applying specific mathematical and spatial algorithms, this workflow inevitably introduces uncertainty into the final model. This inherent uncertainty originates at multiple stages, stemming from both the sparse data inputs and the modelling steps themselves (Høyer et al., 2024) Although this uncertainty is an inescapable aspect of subsurface engineering, precisely measuring its extent is crucial for effective decision-making. This quantification is usually accomplished via Monte Carlo simulations, necessitating the creation of a large collection of equally likely 3D geomodels. Creating this extensive ensemble, however, reveals a core drawback in conventional modeling techniques: the significant compromise between computational efficiency and geological precision. To sustain the generation rate needed for extensive uncertainty quantification, traditional methods frequently depend on greatly simplified, weak geological assumptions.

Due to the fact that these approaches are often limited to simple statistical distributions or strict object-oriented rules (Rongier & Peeters, 2025b), they find it challenging to replicate the intricate, non-linear architectural relationships present in natural sedimentary formations (Sun et al., 2023). As a result, existing workflows compel engineers to decide between models that can quickly quantify uncertainty but do not possess geological realism, or highly precise multi-conceptual models that require too much effort to produce in large quantities (Enemark et al., 2019).This bottleneck highlights a significant deficiency in contemporary subsurface characterization. A significant need exists for a modeling approach that can thoroughly incorporate complex geological prior knowledge while also ensuring the fast generation rates necessary for ensemble-based uncertainty quantification. As a result, the inquiry comes up regarding how the existing modelling process can be modified to ensure that:

- 1) Previous geological insights and structural intricacies are better employed.
- 2) Uncertainty in the produced models is quantified more precisely and efficiently.

To address the need for both geological realism and rapid ensemble generation, recent research has proposed advanced machine learning algorithms as a highly effective solution. These algorithms, specifically the modern subset known as (Deep Generative Modelling (DGM)), possess the unique ability to learn complex, non-linear spatial relationships directly from data. By learning these intricate relationships, DGM can accurately capture the geological complexity and non-stationarity that traditional methods often miss (Rongier & Peeters, 2025b). Furthermore, once such a model is fully trained and validated, it can generate a vast ensemble of geologically realistic 3D realizations almost instantaneously. This rapid generation directly facilitates the large-scale uncertainty quantification demanded by modern subsurface engineering.

1.2. Research gap and opportunity

While the broader application of machine learning in geomodelling has historical roots (Demyanov et al., 2008), recent surges in computing power have yielded highly capable 3D DGM architectures, including Variational Auto Encoders (VAEs) and Latent Diffusion Models (e.g., Bhavsar et al., 2024; Di Federico & Durlofsky, 2025; Laloy et al., 2017; Song et al., 2023). Among these varied architectures, this study specifically focuses on Generative Adversarial Networks (Generative Adversarial Networks (GANs)). While recent literature highlights Latent Diffusion Models as a vanguard for precise hard data conditioning, they inherently suffer from slow inference speeds due to their iterative denoising process. In contrast, GANs offer near-instantaneous sampling speeds perfectly suited for large-scale Monte Carlo uncertainty loops, despite well-known training vulnerabilities such as mode collapse. GANs are therefore selected due to their computational efficiency, established research history in spatial generation, and the wide availability of architectures strictly tailored for geomodelling. However, successfully training such GANs requires an extensive volume of high-quality data to act as the geological prior. To overcome the inherent scarcity of real subsurface data, process-based forward modelling software, such as FLUMY (Martin et al., 2016), can be utilized to generate robust synthetic training datasets.

Utilizing these synthetic datasets allows GANs to effectively learn geological priors; however, a critical research gap remains regarding their ultimate deployment. While DGM excels in theoretical environments, recent literature emphasizes an urgent need to transition from idealized, synthetic test cases to practical field applications. As Rongier and Peeters (2025b) stated, “We now need to move towards more detailed studies on larger case studies closer to field applications to further assess how the whole workflow performs”. Similarly, Di Federico and Durlofsky (2025) noted that the application (and extension as required) of the overall methodology to real cases should be performed. Addressing this explicit mandate forms the core opportunity of this research. Consequently, the primary goal of this project is to bridge the divide between synthetic training environments and practical field operations by assessing the performance of deep generative modelling in a more complex, real-world application.

Testing these models in a real-world application requires targeting an active subsurface system, a role fulfilled in this study by utilizing data from the Delft Sandstone reservoir. This reservoir target must be characterized using a comprehensive geological conceptual model combined with sparse well data. Such sparse well data will serve as the hard conditioning constraints to evaluate how the generative model behaves under tight physical limitations. These limitations, and the resulting fluctuations in uncertainty, will be analyzed by testing the workflow across multiple distinct geological priors. Investigating the interplay between varied priors and sparse conditioning constraints ultimately establishes the framework for the specific objectives of this study.

1.3. Objective and research questions

The main goal of this thesis is to assess the practical feasibility of using a GAN to generate models of a real-world reservoir by conditioning it to well data. The outcome of this thesis may indicate whether GANs can be used to model the subsurface in various settings and if other DGM techniques should be considered. Hence, the main research question of this work is: **To what extent can Deep Generative Modelling (DGM) be used to create geologically plausible realizations of the Delft Sandstone reservoir while conditioned to sparse real-world data?**
To help answer this question, several subquestions are formulated:

1. How should the Delft Sandstone's geology be parameterized in FLUMY to produce a training dataset sufficiently diverse for DGM modeling?
2. What type of GAN architecture best suits facies modelling?
3. How well is the model able to reproduce the complexity inherent in the training data?
4. How well does the optimized model produce geologically plausible structures?
5. How different do models trained on different geological priors react when sparse data is introduced?

1.4. Thesis outline

The remainder of this thesis is structured to systematically address these research objectives. Chapter 2 establishes the theoretical foundation by reviewing the principles of geological modelling, the mechanics of deep generative models, and the specific geological setting of the Delft Sandstone Member. Building upon this theoretical background, Chapter 3 outlines the core workflow, describing the generation of synthetic training datasets using FLUMY and detailing the specific Generative Adversarial Network (GAN) architecture deployed for the reservoir modelling tasks. The empirical outcomes of this workflow are presented in Chapter 4, which evaluates the training process and analyzes the structural and spatial realism of the generated realizations. These results are critically examined in Chapter 5, which contextualizes the findings, addresses the inherent limitations of the proposed approach, and highlights avenues for future research. Finally, Chapter 6 summarizes the main contributions of this thesis and provides definitive answers to the central research questions.

2 Background

This chapter establishes the theoretical and geological foundations upon which this research is built. It provides a comprehensive overview of the geological setting of the Delft reservoir and a literature review of modern geomodelling and machine learning (ML) techniques. Section 2.1 introduces the geological context of the Delft reservoir, detailing its depositional environment and current conceptual model. Section 2.2 explores the state-of-the-art in geological modelling, comparing traditional techniques and their respective limitations. Finally, Section 2.3 provides an in-depth review of machine learning and deep learning (DL) principles, focusing on their historical evolution and contemporary application to subsurface characterization. This balanced overview is intended to provide both geological and computer science readers with the necessary context to understand the research methodology and results presented in subsequent chapters.

2.1. The Delft Geothermal Reservoir

The data utilized in this study is sourced from the Delft geothermal reservoir, situated beneath the Technische Universiteit Delft (TUD) Campus in the Netherlands. As of February 2026, a geothermal doublet is actively producing heat from this reservoir (Beerda, 2026). A doublet system consists of a two-well setup: a producer well extracts warm formation water, and an injector well returns the cooled water to the reservoir. At the Delft site, the producer well (**DEL-GT-01**) was drilled to a True Vertical Depth (TvD) of approximately 2,337 m, while the injector well (**DEL-GT-02-S2**) reached a TvD of approximately 2,100 m (Vardon et al., 2024). The project serves both commercial and research objectives, acquiring extensive geophysical logs and core data, supplemented by surface and downhole monitoring instrumentation (Vardon et al., 2020; Vardon et al., 2024). Despite the high data density, the cored section is restricted to the first 80 m of the **DEL-GT-01** well (approximately one-fourth of the reservoir thickness), while **DEL-GT-02-S2** contains no cored sections. The primary target is the Delft Sandstone (DSST), a member of the Lower Cretaceous Nieuwerkerk Formation in the West Netherlands Basin (WNB) (Donselaar et al., 2015).

2.1.1. Geology of the Delft Reservoir

The WNB is a 60 km wide transtensional basin in the southwest of the Netherlands, characterized by NW-SE trending structural highs and depressions (Donselaar et al., 2015; Ziegler, 1990). The WNB entered an active rifting stage during the Middle Jurassic (Bodenhausen & Ott, 1981; Den Hartog Jager, 1996; Willems et al., 2020), leading to the formation of several half-grabens filled with terrestrial sediments sourced from neighboring regions (Den Hartog Jager, 1996). The progressive filling of these half-grabens coincided with a change in the depositional environment as a transgressive shoreline moved the system from fluvial to marine. The Nieuwerkerk Formation comprises three members: the Alblasserdam, the DSST, and the Rodenrijs Claystone.

The Alblasserdam member was deposited in a fluvial environment characterized by stacked channels and floodplain deposits (DeVault & Jeremiah, 2002). The DSST directly overlays the Alblasserdam and consists of thick, stacked fluvial-deltaic channel sandstones with high net-to-gross values, oriented NW-SE (Bruining, 2023; Groenenberg et al., 2010; Mijnlieff, 2020; Willems et al., 2020). The DSST is in turn overlain by the Rodenrijs Claystone, an organic-rich lagoonal deposit (Bruining, 2023; DeVault & Jeremiah, 2002; Willems et al., 2020).

Based on the sedimentological analysis of the **DEL-GT-01** core and **DEL-GT-02-S2** sidewall samples, three primary facies associations (FAs) have been identified (Bruining, 2023):

- **FA1: Fluvial and distributary channel deposits.** Primarily light grey, fine- to medium-grained sandstones featuring low-angle cross-bedding and ripple lamination, with unit thicknesses ranging from 5 to 80 m.

- **FA2: Crevasse splays and levee deposits.** Composed of moderately sorted, very fine- to medium-grained sandstones interbedded with dark grey siltstones, with thicknesses ranging from 1 to 5 m.

- **FA3: Overbank deposits and paleosols.** Consists of dark grey siltstones, claystones, and very fine-grained sand deposited in low-energy environments. This association is characterized by abundant organic material, slickensides, and siderite nodules.

The specific depositional environment of the DSST remains a subject of scientific discussion. It has been attributed to both fluvial-meandering (Donselaar et al., 2015; Vondrak et al., 2018; Willems et al., 2020) and fluvio-deltaic systems (Bruining, 2023; van Adrichem Boogaert & Kouwe, 1993; Weert et al., 2024). Recent analysis suggests that the cored section represents a deltaic environment (Bruining, 2023). However, due to limited core coverage, it is hypothesized that the lower DSST likely represents a fluvial environment, while the upper section reflects the transgressive shift toward a deltaic system.

2.1.2. Deltaic Depositional Environment

Deltaic environments form at the land-sea interface where river systems supply clastic sediments. Delta morphology is governed by the relative influence of fluvial, wave, and tidal processes (Kolb & Van Lopik, 1966). The subaerial delta plain is typically divided into two domains (Heldreich et al., 2017; Kolb & Van Lopik, 1966):

- **Upper Delta Plain:** Lies above the limit of saltwater intrusion. Sedimentation is driven by distributary channel migration, overbank flooding, and crevassing. Environments include meandering and braided channels, lacustrine fills, and floodplains.

- **Lower Delta Plain:** Subject to river-marine interaction and tidal influence. Channels are more numerous and exhibit bifurcating or anastomosing patterns. Interdistributary bays, bay fills (crevasse splays), and marshes dominate the landscape.

In the DSST, FA1 features organic material and siltstones indicating slackwater periods, while the lack of bioturbation suggests high sedimentation rates. The fine-grained nature of FA2 and FA3 reflects more distal positions on the delta plain. Siderite nodules in FA3 indicate anoxic, waterlogged conditions (Samuel, 1959), while soft-sediment deformation suggests rapid loading of mudstones over saturated sands. These features collectively indicate that FA3 represents deposition within a delta plain, likely in an interdistributary bay or a pro-delta environment.

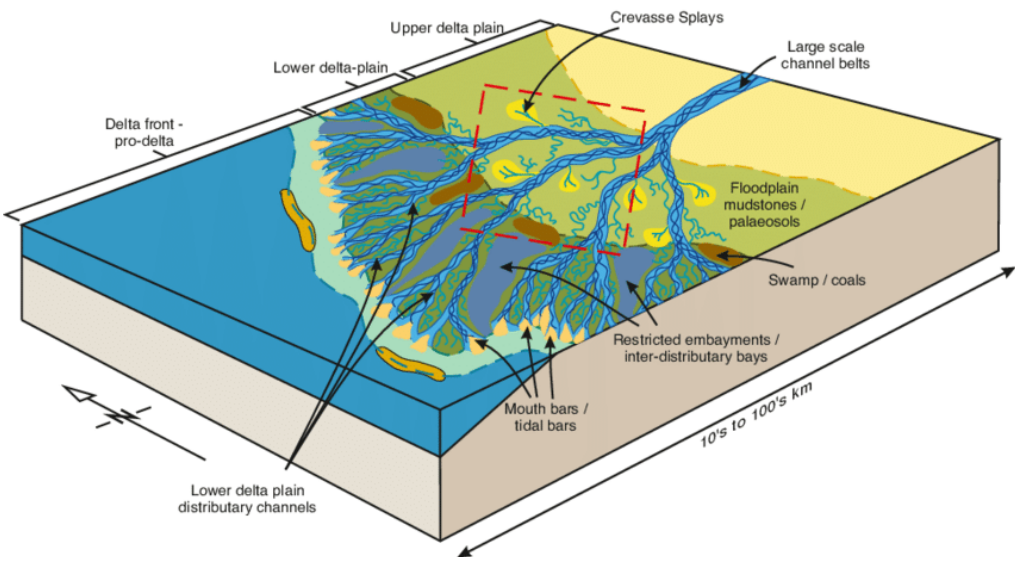

Figure 2.1: Depositional model of the Mungaroo Formation, illustrating the geometries and domains of a fluvial-deltaic system (Heldreich et al., 2017).

2.2. Geological Modelling

The primary objective of 3D reservoir modelling is to provide a representative volume of the subsurface for flow simulation and decision support. These models are dynamic, requiring updates as new data becomes available to refine predictions over the reservoir's lifetime (Emerick, 2025; Ringrose & Bentley, 2016). Existing techniques are generally categorized into three types (Koltermann & Gorelick, 1996):

- **Geostatistical (Pixel-based) Models:** Use spatial statistics (e.g., variograms) for cell-by-cell property assignment. They are efficient and honor dense well data but often fail to capture complex connectivity, resulting in pixelated realizations.
- **Object-based Models:** Directly place geometric features (e.g., channels) based on geological rules. They offer high visual realism but struggle to honor dense or closely spaced well data.
- **Process-based Models:** Simulate the physical laws of sediment transport and deposition (conservation of mass and momentum). They provide the highest degree of realism but are computationally expensive and difficult to condition to specific hard data like well logs.

To leverage the strengths of each, hybrid models have been developed (Lopez et al., 2009; Michael et al., 2010). Table 2.1 summarizes the comparison between these techniques.

Attribute	Geostatistical	Object-Based	Process-Based
Geological Prior	Variograms, training images.	Geometric shapes and rules.	Physical boundary conditions.
Realism	Low to Moderate.	Moderate to High.	Very High.
Key Advantages	Computational speed, well conditioning.	Realistic connectivity.	Physical consistency.
Key Disadvantages	Lack of crisp shapes.	Difficult conditioning.	High computational cost.

Table 2.1: Comparison of spatial modelling techniques in geomodelling.

2.3. Machine Learning

Machine learning (ML) involves the development of algorithms that learn patterns and inference from data to perform specific tasks without explicit programming (Goodfellow et al., 2016). The field has evolved significantly since Samuel (1959) introduced the term, leading to diverse techniques applied across various domains. ML is primarily categorized into:

- **Supervised Learning:** The algorithm is trained on a labeled dataset (input-output pairs).
- **Unsupervised Learning:** The algorithm identifies patterns in unlabeled data without prior knowledge of the output.

Common ML tasks include classification, regression, clustering, and generation. The modern state of ML is largely driven by Artificial Neural Networks (ANNs). Inspired by the biological structure of the brain, ANNs consist of interconnected artificial neurons that process information across multiple layers (Goodfellow et al., 2016). These networks excel at uncovering complex, non-linear relationships in vast datasets (LeCun et al., 2015).

2.3.1. Deep Learning and Inductive Biases

Deep learning (DL) is a subset of ML focusing on multi-layered neural networks that represent data with multiple levels of abstraction (Goodfellow et al., 2016; LeCun et al., 2015). Two critical concepts in DL are the "curse of dimensionality" and "inductive biases." The curse of dimensionality describes the exponential increase in data required as the dimensionality grows, making high-dimensional estimation computationally challenging.

Inductive biases are architectural choices that prioritize certain outcomes, helping models generalize from limited data (Prince, 2023). Different architectures apply specific biases:

- **Convolutional Neural Networks (CNNs):** Leverage spatial locality and translation invariance, making them ideal for image and 3D volume tasks.
- **Recurrent Neural Networks (RNNs):** Designed for sequential data to capture temporal dependencies.
- **Physics-Informed Neural Networks (PINNs):** Incorporate physical laws into the loss function to ensure consistency with scientific principles.

A critical component of the DL workflow is the use of validation and testing protocols to prevent overfitting and ensure that learned patterns are robust and applicable to unseen data.

2.3.2. Geomodelling with Deep Learning

Integration of DL into geomodelling has evolved from simple pixel-based interpolations to sophisticated structural representations. DL allows for the direct learning of geological features from large datasets or process-based simulations, bridging the gap between descriptive statistics and geological realism (Laloy et al., 2017). CNNs have been particularly successful in identifying structural features like faults or channels from seismic and well data (Sun et al., 2023). These techniques enable the generation of stochastic realizations that honor both the global geological concept and local conditioning data, enhancing uncertainty characterization (Song et al., 2023).

2.3.3. Generative Adversarial Networks

Generative Adversarial Networks (GANs) utilize a competitive framework between two networks (Goodfellow et al., 2014): a Generator ($\mathbf{G}$) and a Discriminator ($\mathbf{D}$). The generator creates synthetic samples, while the discriminator attempts to distinguish them from real data. This minimax game is expressed as:

$$\min_{\mathbf{G}} \max_{\mathbf{D}} V(\mathbf{D}, \mathbf{G}) = \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}(\mathbf{x})}[\log \mathbf{D}(\mathbf{x})] + \mathbb{E}_{\mathbf{z} \sim p_{\mathbf{z}}(\mathbf{z})}[\log(1 - \mathbf{D}(\mathbf{G}(\mathbf{z})))] \tag{2.1}$$

where $\mathbf{x}$ represents real data and $\mathbf{z}$ is a latent vector from a prior distribution $p_{\mathbf{z}}$. GANs can learn to generate high-fidelity geological realizations that capture complex features without an explicit likelihood function. Conditional GANs (cGANs) allow auxiliary information (e.g., well logs) to steer the generation process (Mirza & Osindero, 2014).

Recent research has focused on improving the stability of GAN-based geomodels through refined distance metrics like the Mean Sliced Wasserstein Distance (MSWD) (Bhavsar et al., 2024). MSWD projects high-dimensional distributions onto 1D slices, providing a computationally efficient metric for training stability and geological consistency. Furthermore, techniques like latent space optimization and one-hot encoding for categorical facies data have enhanced the reliability of these models in reservoir conditioning tasks (Bhavsar et al., 2024; Rongier & Peeters, 2025b).

3 Methodology

This chapter describes the comprehensive project workflow, beginning with the generation of 3D synthetic datasets used for training, validating, and testing the models (Section 3.1). Section 3.2 details the Generative Adversarial Network (GAN) framework, including the network architectures, the facies representation strategy, and the loss functions employed during training. Finally, Section 3.2.2 introduces the metrics and simulation methods used to evaluate the geological realism and practical utility of the generated realizations.

3.1. Training Data Generation

The training datasets were generated using FLUMY (Martin et al., 2016), a process-based stochastic modelling software designed to reproduce meandering channelized reservoirs. FLUMY has previously successfully been implemented in the research by Bhavsar et al. (2024) to generate samples for training GAN's. FLUMY simulates facies, relative grain size, and depositional time for each voxel in a 3D grid based on specified hydraulic and sedimentological rules.

3.1.1. Conceptual Models

Training deep learning algorithms for 3D geomodelling requires a substantial volume of ground-truth data. Given the scarcity and high uncertainty of subsurface measurements, synthetic data generation using stochastic algorithms is the established approach to provide sufficient training samples (Rongier & Peeters, 2025b; Sun et al., 2023). In this project, three distinct geological conceptual models are employed to capture the range of environments hypothesized for the Delft Sandstone (see Chapter 2):

- **Dataset A (Upper Plain Deltaic):** Characterized by meandering fluvial channels and extensive floodplain deposits.
- **Dataset B (Lower Plain Deltaic):** Reflects increased marine influence with shorter, more numerous distributary channels.
- **Dataset C (General Deltaic):** A stochastic combination of both settings to represent a transitional depositional system.

3.1.2. Dataset Descriptions

The geological variability between the three datasets is controlled by three primary NEXUS input parameters: Net-to-Gross (NTG), maximum channel depth, and maximum sand body extension. The specific configurations are summarized in Table 3.1. The models are defined on a reservoir scale with a grid resolution of $128 \times 128 \times 32$ voxels. With a voxel dimension of 20 m $\times$ 20 m $\times$ 1 m, the total simulated volume covers 2560 m $\times$ 2560 m $\times$ 32 m. While Chapter 2 notes that individual FA1 units can reach 80 m in thickness, the 32 m grid depth used here represents a focused reservoir interval where the conditioning challenge is most acute.

Parameter	Dataset A	Dataset B	Dataset C
Net-to-Gross (%)	67	67	67
Max. Channel Depth (m)	6	4	4–6
Max. Sand Body Extension	100	60	60–100

Table 3.1: NEXUS input parameters for the three training datasets.

Each sample is represented using 9 primary facies classes relevant to the fluvial-deltaic system, as shown in Table 3.2.

Value	Facies Key	Color	Description
0	Background		Undefined/Muddy matrix
1	Channel Lag		Coarse active channel fill
2	Point Bar		Meandering lateral accretion
3	Sand Plug		Fine-grained channel abandonment
4	Crevasse Splay		Proximal high-energy overbank
5	Crevasse Channel		Feeder for splay deposits
6	Levee		Bordering channel ridges
7	Overbank		Vegetated floodplain silt
8	Wetland/Paleosol		Organic-rich anoxic sediments

Table 3.2: Overview of the 9 facies classes used in the training datasets.

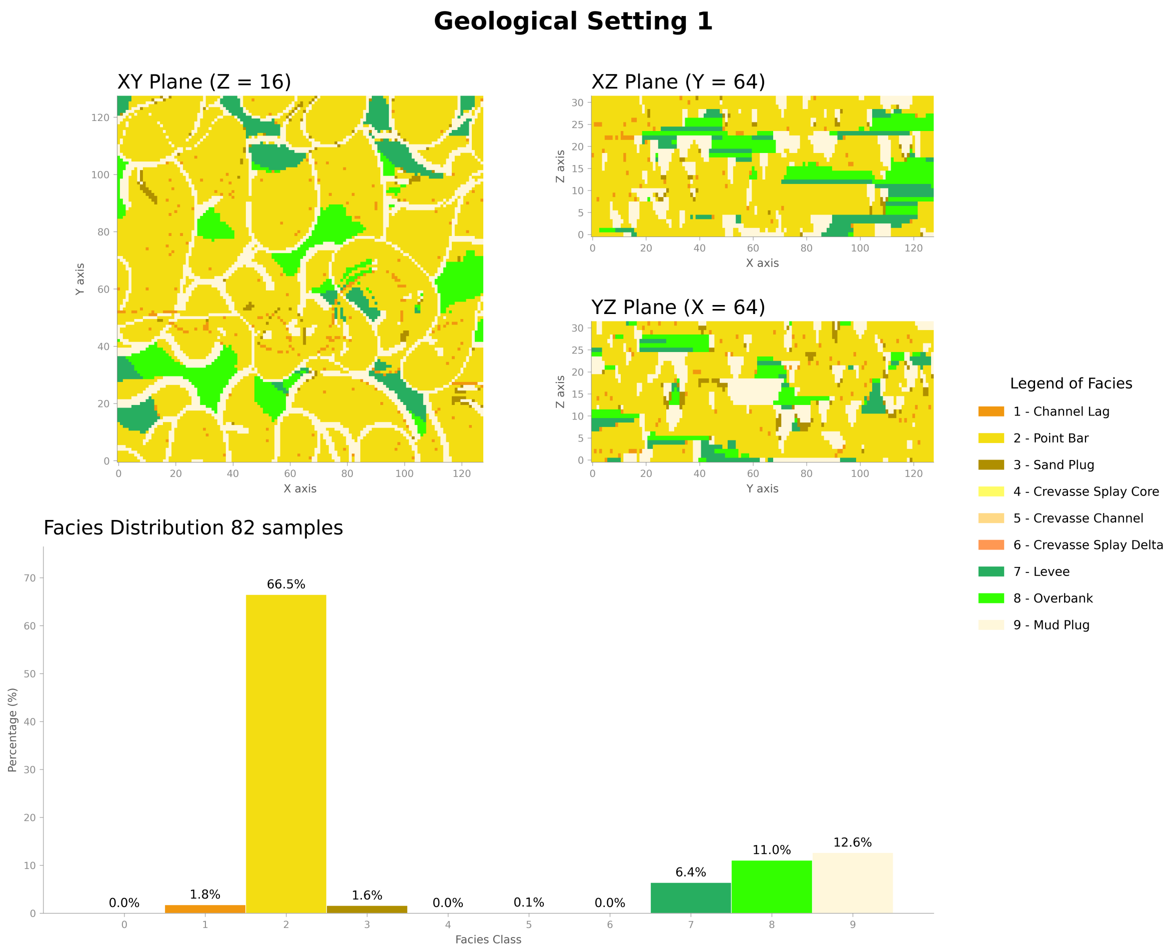

Figure 3.1: Visualization of training data from Dataset A. The rows represent facies distribution, relative grain size, and depositional time, respectively.

3.2. Fluvgan Framework

The Fluvial Generative Adversarial Network (FluvGan) model, introduced by Rongier and Peeters (2025a), is a conditional GAN architecture optimized for 3D fluvial facies modelling. It builds upon the BigGAN framework (Brock et al., 2019) to handle the high resolution and complex connectivity required for reservoir models. Previous studies have demonstrated that FluvGan can generate geologically plausible realizations while remaining training-stable (Rongier & Peeters, 2025b). Crucially, Rongier and Peeters (2025c) showed that the model's latent space can be inverted and optimized to honor hard constraints from well data.

3.2.1. Model Architecture

The architecture comprises a Generator (**G**) and a Discriminator (**D**) using 3D Residual blocks. To handle the categorical nature of facies data, the project employs a one-hot encoding strategy where each facies is represented as a separate channel in the output volume. The generator transforms a 100-dimensional latent vector $\mathbf{z}$ into a $9 \times 32 \times 128 \times 128$ volume. It utilizes a 3D transposed convolution to project the latent vector into a $1024 \times 4 \times 4 \times 4$ feature map, followed by five residual upscaling blocks. To accommodate the anisotropy of the grid, spatial dimensions are doubled in each block (up to 128), while the depth dimension is doubled only in the first three blocks (up to 32). The final layer uses a Softmax activation to ensure the 9 facies channels sum to one per voxel. The discriminator acts as a binary classifier, mapping the $9 \times 32 \times 128 \times 128$ input back to a single scalar logit. It consists of five residual downscaling blocks that iteratively reduce the spatial resolution while increasing the channel count to 1024. Both networks employ spectral normalization and LeakyReLU activations to ensure stable training. The generator and discriminator architectures contain approximately 11.6 and 2.0 million trainable parameters, respectively.

Hyperparameter	**Value**	**Description**
Latent Dimension ($dim(\mathbf{z})$)	100	Gaussian prior
Batch Size	16	Samples per iteration
Learning Rate (G, D)	$10^{-4}, 4 \cdot 10^{-4}$	Two-timescale update rule
Kernel Size	$3 \times 3 \times 3$	3D Convolutional filters
Optimizer	Adam	$\beta_1 = 0, \beta_2 = 0.999$

Table 3.3: Key hyperparameters for the Fluvgan architecture.

3.2.2. Conditioning and Evaluation

To honor sparse well data, a latent space optimization approach is used. The latent vector $\mathbf{z}$ is iteratively refined to minimize the difference between the generated facies at well locations and the true observations. The realism of the final realizations is evaluated using the Mean Sliced Wasserstein Distance (MSWD) across 1,000 projections, ensuring that the multi-point statistics and connectivity of the training data are preserved (Bhavsar et al., 2024).

4 Results

4.1. Post Training Results

4.2. Post Conditioning Results

5 Discussion & Recommendations

5.1. Interpretations of key results

5.2. Limitations of the current work

5.3. Recommendations for future work

6 Conclusion

6.1. Answers to Research Questions

6.2. Final Remarks

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